

MA269 Mock 2 - Model Solutions

① a) i) Asymptotic series $\Leftrightarrow \forall M \leq N, \forall x,$
$$\frac{f(x, \varepsilon) - \sum_{n=0}^M f_n(x, \varepsilon)}{f_M(x, \varepsilon)} \rightarrow 0 \quad \text{as } \varepsilon \rightarrow 0 \quad [2]$$

ii) Poincaré $\Leftrightarrow \forall n, f_n(x, \varepsilon) = a_n(x) \alpha_n(\varepsilon) \quad [1]$

iii) Given $\alpha_n(\varepsilon)$, then

$$\frac{f(x, \varepsilon) - \sum_{j=0}^{n-1} a_j(x) \alpha_j(\varepsilon) - a_n(x) \alpha_n(\varepsilon)}{a_n(x) \alpha_n(\varepsilon)} \rightarrow 0$$

$$\therefore \frac{f(x, \varepsilon) - \sum_{j=0}^{n-1} a_j(x) \alpha_j(\varepsilon)}{\alpha_n(\varepsilon)} \rightarrow a_n(x)$$

So can calculate $a_0(x)$, then $a_1(x)$, then $a_2(x), \dots \quad [4]$

b) [Thinking ... $\tanh x \rightarrow 1$ as $x \rightarrow \infty$, so $x \rightarrow \infty$ as $\varepsilon \rightarrow 0$]

$$\begin{aligned} \tanh x &= \frac{e^x - e^{-x}}{e^x + e^{-x}} = \frac{1 - e^{-2x}}{1 + e^{-2x}} = (1 - e^{-2x})(1 - e^{-2x} + (e^{-2x})^2 + \dots) \\ &= 1 - 2e^{-2x} + 2(e^{-2x})^2 + \dots = 1 - \varepsilon \end{aligned}$$

To leading order, try $-2e^{-2x_0} = -\varepsilon$
 $\Rightarrow x_0 = \frac{1}{2} \log \frac{2}{\varepsilon} = \frac{1}{2} \log \frac{1}{\varepsilon} + \frac{1}{2} \log 2$

2)

Try $x = x_0 + e_0$ with $e_0 = o(x_0)$: (investigate next order)

$$\begin{aligned} \tanh(x_0 + e_0) &= 1 - 2e^{-2x_0} e^{-2e_0} + 2(e^{-2x_0})^2 + \dots \\ &= 1 - \epsilon e^{-2e_0} + \frac{1}{2}\epsilon^2 + \dots \\ &= 1 - \epsilon + 2\epsilon e_0 + \dots + \frac{1}{2}\epsilon^2 + \dots = 1 - \epsilon \\ \Rightarrow e_0 &= -\frac{1}{4}\epsilon \quad (e_0 = O(\epsilon)) \end{aligned}$$

(calculating next order)

$$\therefore x \sim \underbrace{\frac{1}{2} \log \frac{1}{\epsilon} + \frac{1}{2} \log 2}_{\text{first two terms (depending on expansion)}} - \frac{1}{4}\epsilon \sim \underbrace{\frac{1}{2} \log \left(\frac{2}{\epsilon} \right) - \frac{1}{4}\epsilon}_{[9]}$$

c) $f(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} \frac{-5i e^{i\omega t}}{(\omega+i)(\omega^2-2i\omega-2)} d\omega$

Poles when $\omega = -i$ or $\omega = i \pm \sqrt{i^2+2} = \pm 1+i$

$$\text{Res}(-i) = \frac{-5i e^{i(-i)t}}{(-i)^2 - 2i(-i) - 2} = \frac{-5i e^t}{-1-2-2} = i e^t$$

$$\begin{aligned} \text{Res}(\pm 1+i) &= \frac{-5i e^{\pm it} e^{-t}}{(\pm 1+2i)(\pm 1+i - (\mp 1+i))} = \frac{\mp \frac{5}{2} i e^{\pm it} e^{-t} (\pm 1-2i)}{(1^2+2^2)} \\ &= \left(\mp 1 - \frac{1}{2}i \right) e^{\pm it} e^{-t} \end{aligned}$$

For $t < 0$, close contour in Lower Half Plane, to get

$$f(t) = \frac{1}{2\pi} \cdot (-2\pi i) \cdot \text{Res}(-i) = -i \cdot i e^t = e^t$$

(go round pole in clockwise direction)

3)

For $t > 0$, close contour in Upper Half Plane, to get ✓

$$\begin{aligned}
 f(t) &= \frac{1}{2\pi} \cdot 2\pi i \cdot (\text{Res}(1+i) + \text{Res}(-1+i)) \quad \checkmark \\
 &= i \left[\left(-1 - \frac{1}{2}i\right) e^{it} e^{-t} + \left(1 - \frac{1}{2}i\right) e^{-it} e^{-t} \right] \\
 &= e^{-t} \left[\frac{1}{2} (e^{it} + e^{-it}) - i (e^{it} - e^{-it}) \right] \quad \checkmark \\
 &= e^{-t} (\cos t + 2 \sin t)
 \end{aligned}$$

so
$$f(t) = \begin{cases} e^t & t < 0 \\ e^{-t} (\cos t + 2 \sin t) & t > 0 \end{cases}$$

[16]

d)
$$\widehat{f(t-a)} = \hat{f}(s) e^{-sa} \quad \checkmark$$

$$\widehat{\dot{f}(t)} = -f(0) + s \hat{f}(s) \quad \checkmark$$

$$\widehat{t f(t)} = -\frac{d\hat{f}}{ds} \quad \checkmark$$

$$\therefore \hat{g} = -\frac{d}{ds} \left(-f(0) + s \hat{f}(s) e^{-sa} \right)$$

(Note: could be $-f(-a)$, but doesn't affect answer)

$$= -\frac{d}{ds} \left(s e^{-sa} \hat{f}(s) \right) = -e^{-sa} \hat{f}(s) + a s e^{-sa} \hat{f}(s) - s e^{-sa} \frac{d\hat{f}}{ds}.$$

[8]

4)

$$\textcircled{2} \text{ a) i) } \tanh x = \frac{e^x - e^{-x}}{e^x + e^{-x}} = \frac{1 - e^{-2x}}{1 + e^{-2x}} = (1 - e^{-2x})(1 - e^{-2x} + (e^{-2x})^2 + \dots)$$

$$= 1 - 2e^{-2x} + 2(e^{-2x})^2 + \dots = 1 + O(e^{-2x}). \quad [2]$$

$$\text{ii) } \log(\cosh x) = \log\left(\frac{1}{2}(e^x + e^{-x})\right) = \log\left(\frac{e^x}{2}(1 + e^{-2x})\right)$$

$$= x - \log 2 + \log(1 + e^{-2x}) = x - \log 2 + \underbrace{e^{-2x} - \frac{1}{2}(e^{-2x})^2 + \dots}_{O(e^{-2x})} \quad [2]$$

(change var)
 $x = \sqrt{\epsilon} \chi$

(split)

$$\text{b) } I(\epsilon) = \int_0^a \frac{\tanh x}{1 + \epsilon x^2} dx + \int_{a\sqrt{\epsilon}}^{\infty} \frac{\tanh(x/\sqrt{\epsilon})}{1 + \chi^2} \frac{d\chi}{\sqrt{\epsilon}}$$

$$= \int_0^a \tanh x (1 - \epsilon x^2 + \epsilon^2 x^4 + \dots) dx$$

$$+ \frac{1}{\sqrt{\epsilon}} \int_{a\sqrt{\epsilon}}^{\infty} \frac{1 + O(e^{-2x/\sqrt{\epsilon}})}{1 + \chi^2} d\chi \quad \leftarrow \text{exponentially small}$$

$$= \left[\log(\cosh x) \right]_0^a + O(\epsilon a^3) + \frac{1}{\sqrt{\epsilon}} \left[\theta \right]_{\tan^{-1}(a\sqrt{\epsilon})}^{\pi/2} + O(e^{-2a})$$

need a large, ϵa^3 small. e.g. $a = \epsilon^{-1/6}$. (some statement about a large)

$$= \log(\cosh a) + \frac{\pi}{2\sqrt{\epsilon}} - \frac{1}{\sqrt{\epsilon}} \tan^{-1}(a\sqrt{\epsilon}) + O(\epsilon a^3, e^{-2a})$$

$$= \cancel{a} - \log 2 + O(e^{-2a}) + \frac{\pi}{2\sqrt{\epsilon}} - \cancel{a} + O(a^3 \epsilon)$$

$$= \frac{\pi}{2\sqrt{\epsilon}} - \log 2 + o(1).$$

(some analysis of errors, so we're sure we've got first two terms) [16]

5)

3) a) $\frac{dy_0}{dx} - y_0 \cos x = 0 \Rightarrow \frac{1}{y_0} \frac{dy_0}{dx} = \cos x$
 $\Rightarrow \log y_0 = \sin x + C \Rightarrow y_0 = A e^{\sin x} = -e^{\sin x}$ (boundary condition) [4]

b) Include $\varepsilon \frac{d^2 y}{dx^2}$. $x = \varepsilon^\alpha \xi$,

$$\varepsilon^{1-2\alpha} \frac{d^2 y}{d\xi^2} + \varepsilon^{-\alpha} \frac{dy}{d\xi} - y \cos(\varepsilon^\alpha \xi) = 0$$

balance: $1-2\alpha = -\alpha \Rightarrow \alpha = 1$.

$$\therefore \frac{d^2 y}{d\xi^2} + \frac{dy}{d\xi} - \varepsilon y \cos(\varepsilon \xi) = 0$$

boundary condition: $\frac{d^2 y_0}{d\xi^2} + \frac{dy_0}{d\xi} = 0 \Rightarrow y_0 = B + D e^{-\xi} = B + (1-B) e^{-\xi}$ [7]

c) Inner expansion of outer: $y_0 = -e^{\sin(\varepsilon \xi)} = -1 + O(\varepsilon)$

Outer expansion of inner: $y_0 = B + (1-B) e^{-x/\varepsilon} = B + O(e^{-x/\varepsilon})$

Equating $\Rightarrow B = -1$ [6]

d) Additive composite expansion:

$$y_0 = \underbrace{-e^{-\sin x}}_{\text{outer}} + \underbrace{-1 + 2e^{-x/\varepsilon}}_{\text{inner}} - (-1)_{\text{common}}$$

$$= 2e^{-x/\varepsilon} - e^{-\sin x}$$
 [3]

6)

④ a) F.T. $x \mapsto k$

$$\frac{\partial^2 \tilde{u}}{\partial t^2} + \underbrace{(ik)^4}_{k^4} \tilde{u} = \tilde{F}(k), \quad \tilde{u}(k, 0) = \frac{\partial \tilde{u}}{\partial t}(k, 0) = 0$$

Solve: $\tilde{u} = \underbrace{A e^{ik^2 t} + B e^{-ik^2 t}}_{\text{C.F.}} + \frac{\tilde{F}(k)}{k^4} \leftarrow \text{P.I.}$

B.c.s: $\tilde{u}(k, 0) = A + B + \frac{\tilde{F}}{k^4} = 0$

$$\frac{\partial \tilde{u}}{\partial t}(k, 0) = ik^2(A - B) = 0 \Rightarrow A = B = -\frac{\tilde{F}}{2k^4}$$

$$\therefore \tilde{u}(k, t) = -\frac{\tilde{F}}{2k^4} e^{ik^2 t} - \frac{\tilde{F}}{2k^4} e^{-ik^2 t} + \frac{\tilde{F}}{k^4} = \frac{\tilde{F}}{k^4} (1 - \cos(k^2 t))$$

so $u(x, t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} \frac{1 - \cos(k^2 t)}{k^4} \tilde{F}(k) e^{ikx} dk.$

[14]

b) i) As $k \rightarrow 0$, $\frac{1 - \cos(k^2 t)}{k^4} \sim \frac{1 - \left(1 - \frac{1}{2}(k^2 t)^2 + \frac{1}{4!}(k^2 t)^4 + \dots\right)}{k^4}$

$$= \frac{1}{2} t^2 + O(k^4)$$

So removable singularity at $k=0$.

ii) As $|k| \rightarrow \infty$, $\cos(k^2 t) = \frac{1}{2} (e^{ik^2 t} + e^{-ik^2 t})$ will be exponentially large, so can't close in upper half-plane or lower half-plane.

[6]

(See, e.g. Mock 1, Q.2. for what to do if you find this in practical problems.)

7)

⑤ a) $\int_0^t F(t-\tau) G(\tau) d\tau$ is Laplace convolution, so use Laplace transform.

$$\widehat{t f(t)} = -\frac{d\hat{f}}{ds}, \quad \hat{1} = \frac{1}{s} \Rightarrow \hat{t^2} = \frac{d^2}{ds^2} \left(\frac{1}{s} \right) = \frac{2}{s^3}.$$

$$\widehat{e^{-at}} = \int_0^\infty e^{-at} e^{-st} dt = \left[\frac{e^{-(a+s)t}}{-(a+s)} \right]_0^\infty = \frac{1}{s+a}$$

$$\widehat{\frac{df}{dt}} = -f(0) + s\hat{f}(s), \quad \text{but } f(0) = 0 \text{ here.}$$

$$\text{so } \hat{f}(s) = \frac{2}{s^3} + s\hat{f}(s) \frac{1}{s+a} \Rightarrow \left(\frac{s+a}{s+a} - \frac{s}{s+a} \right) \hat{f} = \frac{2}{s^3}$$

$$\Rightarrow \hat{f} = \frac{2}{a} \frac{s+a}{s^3} = \frac{1}{a} \frac{2}{s^2} + 2 \frac{1}{s^3}$$

$$\therefore f(t) = \frac{2}{a} t + t^2 \quad (\text{by inspection})$$

[16]

$$\text{b) If } f(0) = b \neq 0, \text{ then } \hat{f} = \frac{2}{s^3} + s\hat{f} \frac{1}{s+a} - \frac{b}{s+a},$$

$$\Rightarrow \hat{f} = \frac{1}{a} \frac{2}{s^3} + 2 \frac{1}{s^2} - \frac{b}{a}$$

But new term $\rightarrow 0$ as $s \rightarrow \infty$, so must be singular at $t=0$. Therefore, no such solution.

[4]